§3.3 Diffusion with a source

Diffusion with a source on the whole line

Consider the inhomogeneous diffusion equation on the whole line,

$$\begin{cases} u_t - ku_{xx} = f(x,t), & -\infty < x < \infty, \quad t > 0, \\ u(x,0) = \phi(x) & \text{(initial condition at } t = 0) \end{cases} \tag{35}$$

with $f(x,t)$ and $\phi(x)$ arbitrary given functions.

$u_1(x,t)$ and $u_2(x,t)$ solutions

$v(x,t) = u_1(x,t) - u_2(x,t)$ solves

$$\begin{cases} v_t - kv_{xx} = 0 \\ v(x,0) = 0 \end{cases}$$

§3.3 Diffusion with a source

Diffusion with a source on the whole line

Consider the inhomogeneous diffusion equation on the whole line,

$$\begin{aligned} &u_t - ku_{xx} = f(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = \phi(x) \quad \text{(initial condition at } t = 0) \end{aligned} \tag{35}$$

with $f(x,t)$ and $\phi(x)$ arbitrary given functions.

- The right hand side of the equation, $f(x,t)$, is called the source term.

§3.3 Diffusion with a source

Diffusion with a source on the whole line

Consider the inhomogeneous diffusion equation on the whole line,

$$\begin{aligned} &u_t - ku_{xx} = f(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = \phi(x) \quad \text{(initial condition at } t = 0) \end{aligned} \tag{35}$$

with $f(x,t)$ and $\phi(x)$ arbitrary given functions.

- The right hand side of the equation, $f(x,t)$, is called the source term.
- In terms of heat conduction, the source term measures the physical effect of an external heat source. It has units of heat flux (left hand side of the equation has the units of u_t, i.e. change in temperature per unit time), thus it gives the instantaneous temperature change due to an external heat source.

has the same unit as u_t

§3.3 Diffusion with a source

$$u(x,0) = \int_{-\infty}^{\infty} S(x-y,0)\,\phi(y)\,dy$$

Theorem 1. The solution of (35) is,

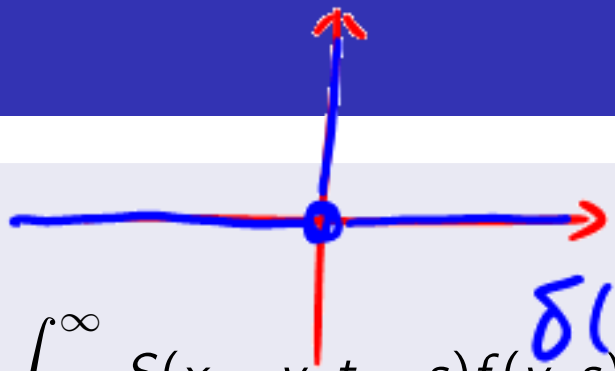
$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds, \quad (36)$$

where $S(x,t)$ is the Gaussian kernel $= \int_{-\infty}^{\infty} \delta(x-y)\,\phi(y)\,dy$

$$S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}. \quad (37)$$

$$\begin{cases} \frac{\partial}{\partial t} S(x,t) - k \frac{\partial^2}{\partial x^2} S(x,t) = 0 \\ S(x,0) = \delta(x) \end{cases}$$

§3.3 Diffusion with a source



$\delta(x) = H'(x)$

Theorem 1. The solution of (35) is,

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds, \quad (36)$$

where $S(x,t)$ is the Gaussian kernel

$$S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}. \quad (37)$$

- In terms of heat conduction, the physical meaning of the expression $\int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy$ in (36) is that the heat kernel averages out the initial temperature distribution along the entire rod.

$$u(x,0) = \int_{-\infty}^{\infty} \underbrace{\delta(x-y)\,\phi(y)}\,dy = \phi(x)$$

when $y = x$, $x - y = 0$ ↳ zero almost everywhere except at $y = x$

§3.3 Diffusion with a source

Theorem 1. The solution of (35) is,

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds, \quad (36)$$

where $S(x,t)$ is the Gaussian kernel

$$S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}. \quad (37)$$

- In terms of heat conduction, the physical meaning of the expression $\int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy$ in (36) is that the heat kernel averages out the initial temperature distribution along the entire rod.
- $f(x,t)$ is the external heat source. It must be averaged out, too; not only over the entire rod, but over time as well, since the heat contribution at an earlier time will affect the temperatures at all later times.

§3.3 Diffusion with a source

Theorem 1. The solution of (35) is,

$g(x,t,s)$

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dy\,ds, \quad (36)$$

where $S(x,t)$ is the Gaussian kernel

$$S(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}. \quad (37)$$

$\frac{\partial}{\partial t}\int_0^t g(x,t,s)ds$

$= g(x,t,t) + \int_0^t g_t ds$

- In terms of heat conduction, the physical meaning of the expression $\int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy$ in (36) is that the heat kernel averages out the initial temperature distribution along the entire rod.
- $f(x,t)$ is the external heat source. It must be averaged out, too; not only over the entire rod, but over time as well, since the heat contribution at an earlier time will affect the temperatures at all later times.
- The time integration is only over the interval $[0,t]$, since the heat contribution at later times can not affect the temperature at time t.

Proof of Theorem 1. We show that (36) indeed solves problem (35) by a direct substitution.

- Differentiate (36) with respect to t, we have

$$\frac{\partial}{\partial t} S(x-y,t) = k \frac{\partial^2}{\partial x^2} S(x-y,t)$$

$$\begin{aligned}
\frac{\partial}{\partial t} u(x,t) &= \frac{\partial}{\partial t} \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy \\
&\quad + \frac{\partial}{\partial t} \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds \\
&= \int_{-\infty}^{\infty} \frac{\partial}{\partial t} S(x-y,t)\phi(y)dy + \int_{-\infty}^{\infty} \underline{\underline{S(x-y,0)}} f(y,t)dy \\
&\quad + \int_0^t \int_{-\infty}^{\infty} \frac{\partial}{\partial t} S(x-y,t-s)f(y,s)dyds
\end{aligned} \tag{38}$$

$$\delta(x-y)$$

$$\int_{-\infty}^{\infty} \delta(x-y) f(y,t)\, dy = f(x,t)$$

§3.3 Diffusion with a source

Proof of Theorem 1. We show that (36) indeed solves problem (35) by a direct substitution.

- Differentiate (36) with respect to t, we have

$$\begin{aligned}\frac{\partial}{\partial t}u(x,t) &= \frac{\partial}{\partial t}\int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy \\ &\quad + \frac{\partial}{\partial t}\int_0^t\int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds \\ &= \int_{-\infty}^{\infty}\frac{\partial}{\partial t}S(x-y,t)\phi(y)dy + \int_{-\infty}^{\infty} S(x-y,0)f(y,t)dy \\ &\quad + \int_0^t\int_{-\infty}^{\infty}\frac{\partial}{\partial t}S(x-y,t-s)f(y,s)dyds\end{aligned} \tag{38}$$

- Recall that Gaussian kernel $S(x,t)$ solves the diffusion equation and has the Dirac delta function as its initial data, i.e. $S_t = kS_{xx}$, and $S(x-y,0) = \delta(x-y)$.

Proof of Theorem 1 (Cont'd).

- Hence,

$$
\begin{aligned}
\frac{\partial}{\partial t}u(x,t) &= \int_{-\infty}^{\infty} k\frac{\partial^2}{\partial x^2}S(x-y,t)\phi(y)dy + \int_{-\infty}^{\infty}\delta(x-y)f(y,t)dy \\
&\quad + \int_0^t\int_{-\infty}^{\infty} k\frac{\partial^2}{\partial x^2}S(x-y,t-s)f(y,s)dyds \\
&= k\frac{\partial^2}{\partial x^2}\int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + f(x,t) \qquad (39) \\
&\quad + k\frac{\partial^2}{\partial x^2}\int_0^t\int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds \\
&= k\frac{\partial^2}{\partial x^2}u(x,t) + f(x,t).
\end{aligned}
$$

Proof of Theorem 1 (Cont'd).

- Hence,

$$\begin{aligned}\frac{\partial}{\partial t}u(x,t) &= \int_{-\infty}^{\infty} k\frac{\partial^2}{\partial x^2}S(x-y,t)\phi(y)dy + \int_{-\infty}^{\infty}\delta(x-y)f(y,t)dy \\ &\quad + \int_0^t\int_{-\infty}^{\infty} k\frac{\partial^2}{\partial x^2}S(x-y,t-s)f(y,s)dyds \\ &= k\frac{\partial^2}{\partial x^2}\int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + f(x,t) \\ &\quad + k\frac{\partial^2}{\partial x^2}\int_0^t\int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds \\ &= k\frac{\partial^2}{\partial x^2}u(x,t) + f(x,t).\end{aligned} \tag{39}$$

- Expression (39) indicates that $u(x,t)$ defined in (36) solves the inhomogeneous diffusion equation $u_t - ku_{xx} = f(x,t)$.

Proof of Theorem 1 (Cont'd).

- We verify the initial condition. The solution is

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds.$$

Letting $t \to 0$, the first term becomes

$$\int_{-\infty}^{\infty} S(x-y,0)\phi(y)dy = \int_{-\infty}^{\infty} \delta(x-y)\phi(y)dy = \phi(x).$$

The second term is an integral from 0 to 0. Therefore, $u(x,0) = \phi(x)$ satisfies the initial condition. This proves that (36) is the unique solution of the IVP (35).

Proof of Theorem 1 (Cont'd).

- We verify the initial condition. The solution is

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy + \int_0^t \int_{-\infty}^{\infty} S(x-y,t-s)f(y,s)dyds.$$

Letting $t \to 0$, the first term becomes

$$\int_{-\infty}^{\infty} S(x-y,0)\phi(y)dy = \int_{-\infty}^{\infty} \delta(x-y)\phi(y)dy = \phi(x).$$

The second term is an integral from 0 to 0. Therefore, $u(x,0) = \phi(x)$ satisfies the initial condition. This proves that (36) is the unique solution of the IVP (35).

- The solution has an explicit form as

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4kt}} \phi(y)dy + \int_0^t \int_{-\infty}^{\infty} \frac{e^{-\frac{(x-y)^2}{4k(t-s)}}}{\sqrt{4\pi k(t-s)}} f(y,s)dyds. \tag{40}$$

Example A (not in the textbook)

Find the solution of the inhomogeneous problem,

$$\begin{aligned} &u_t - ku_{xx} = \delta(x-1)\delta(t-2), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = 0, \qquad t > 0. \end{aligned} \tag{41}$$

$$\int_0^t \int_{-\infty}^{\infty} S(x-y, t-s) f(y,s)\, dy\, ds$$

$$= \int_0^t \int_{-\infty}^{\infty} S(x-y, t-s)\, \delta(y-1)\, \delta(s-2)\, dy\, ds$$

$$= \int_0^t S(x-1, t-s)\, \delta(s-2)\, ds$$

Example A (not in the textbook)

Find the solution of the inhomogeneous problem,

$$\begin{aligned} &u_t - ku_{xx} = \delta(x-1)\delta(t-2), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = 0, \qquad t > 0. \end{aligned} \tag{41}$$

Solution. Using formula (36), and substituting the expression for $f(x,t) = \delta(x-1)\delta(t-2)$, and $\phi(x) = 0$, we get

$$\begin{aligned} u(x,t) &= \int_0^t \int_{-\infty}^{\infty} S(x-y, t-s)\delta(y-1)\delta(s-2)dyds \\ &= \int_0^t S(x-1, t-s)\delta(s-2)ds. \end{aligned}$$

Example A (not in the textbook)

Find the solution of the inhomogeneous problem,

$$\begin{aligned} &u_t - ku_{xx} = \delta(x-1)\delta(t-2), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = 0, \qquad t > 0. \end{aligned} \tag{41}$$

Solution. Using formula (36), and substituting the expression for $f(x,t) = \delta(x-1)\delta(t-2)$, and $\phi(x) = 0$, we get

$$\begin{aligned} u(x,t) &= \int_0^t \int_{-\infty}^{\infty} S(x-y, t-s)\delta(y-1)\delta(s-2)dyds \\ &= \int_0^t S(x-1, t-s)\delta(s-2)ds. \end{aligned}$$

Notice that if $t < 2$, then $\delta(s-2) = 0$ for all $s \in [0,t]$; and if $t > 2$, then the delta function will act on Gaussian kernel by assigning its value at $s = 2$.

Solution (Cont'd). Hence,

$$u(x,t) = \begin{cases} 0, & \text{for } \ 0 < t < 2, \\ S(x-1, t-2), & \text{for } \ t > 2. \end{cases}$$

$$S(x, t)$$

$$\begin{cases} \frac{\partial}{\partial t} S(x,t) - k \frac{\partial^2}{\partial x^2} S(x,t) = 0 \\ S(x, 0) = \delta(x) \end{cases}$$

Solution (Cont'd). Hence,

$$u(x,t) = \begin{cases} 0, & \text{for } \ 0 < t < 2, \\ S(x-1, t-2), & \text{for } \ t > 2. \end{cases}$$

The external heat source in this case gives an instantaneous temperature boost to the point $x = 1$ at time $t = 2$. Henceforth, the temperature in the rod will remain zero till the time $t = 2$, and afterward the heat will transfer exactly as in the case of the homogeneous heat equation with data given at time $t = 2$ as $u(x, 2) = \delta(x - 1)$.

§3.3 Diffusion with a source

Diffusion with a source on a half-line

Consider the more complicated Dirichlet problem of a boundary source $h(t)$ on the half-line; that is,

$$\begin{aligned} &v_t - kv_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &v(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &v(x,0) = \phi(x) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{42}$$

§3.3 Diffusion with a source

Diffusion with a source on a half-line

Consider the more complicated Dirichlet problem of a boundary source $h(t)$ on the half-line; that is,

$$\begin{aligned} &v_t - kv_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &v(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &v(x,0) = \phi(x) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{42}$$

- For inhomogeneous diffusion on the half-line we can use the method of reflection as well.

§3.3 Diffusion with a source

Diffusion with a source on a half-line

Consider the more complicated Dirichlet problem of a boundary source $h(t)$ on the half-line; that is,

$$\begin{aligned} &v_t - kv_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &v(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &v(x,0) = \phi(x) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{42}$$

- For inhomogeneous diffusion on the half-line we can use the method of reflection as well.
- We may use the subtraction device to reduce (42) to a simpler problem with a homogeneous boundary condition. Let

$$V(x,t) = v(x,t) - h(t). \tag{43}$$

$V(0,t) = v(0,t) - h(t) = h(t) - h(t) = 0$

Diffusion with a source on a half-line (Cont'd)

- We can check that $V(x,t) = v(x,t) - h(t)$ satisfies

IBVP

$$\begin{cases} V_t - kV_{xx} = f(x,t) - h'(t), \quad 0 < x < \infty, \quad t > 0, \\ V(0,t) = 0 \quad \text{(boundary condition at } x = 0), \\ V(x,0) = \phi(x) - h(0) \quad \text{(initial condition at } t = 0). \end{cases} \tag{44}$$

This new problem (44) has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions.

Diffusion with a source on a half-line (Cont'd)

- We can check that $V(x,t) = v(x,t) - h(t)$ satisfies

$$\begin{aligned} &V_t - kV_{xx} = f(x,t) - h'(t), \quad 0 < x < \infty, \quad t > 0, \\ &V(0,t) = 0 \quad \text{(boundary condition at } x = 0), \\ &V(x,0) = \phi(x) - h(0) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{44}$$

  This new problem (44) has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions.

- Once we find $V(x,t)$, we can recover $v(x,t)$ by $v(x,t) = V(x,t) + h(t)$.

Diffusion with a source on a half-line (Cont'd)

- We can check that $V(x,t) = v(x,t) - h(t)$ satisfies

$$\begin{aligned} &V_t - kV_{xx} = f(x,t) - h'(t), \quad 0 < x < \infty, \quad t > 0, \\ &V(0,t) = 0 \quad \text{(boundary condition at } x = 0), \\ &V(x,0) = \phi(x) - h(0) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{44}$$

  This new problem (44) has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions.

- Once we find $V(x,t)$, we can recover $v(x,t)$ by $v(x,t) = V(x,t) + h(t)$.
- The domain of independent variables (x,t) in this case is a quarter-plane with specified conditions on both of its half-lines. If they do not agree at the corner [i.e., if $\phi(0) \neq h(0)$], then the solution is discontinuous there (but continuous everywhere else). This is physically sensible. Think for instance, of suddenly at $t = 0$ sticking a hot iron bar into a cold bath.

Diffusion with a source on a half-line (Cont'd)

For the inhomogeneous Neumann problem on the half line

$$\begin{aligned} &w_t - kw_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &w_x(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &w(x,0) = \phi(x) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{45}$$

Diffusion with a source on a half-line (Cont'd)

For the inhomogeneous Neumann problem on the half line

$$\begin{aligned} &w_t - kw_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &w_x(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &w(x,0) = \phi(x) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{45}$$

- Let $W(x,t) = w(x,t) - xh(t)$, we can check that $W(x,t)$ satisfies

IBVP

$$\begin{cases} W_t - kW_{xx} = f(x,t) - xh'(t), \quad 0 < x < \infty, \quad t > 0, \\ W_x(0,t) = 0 \quad \text{(boundary condition at } x = 0), \\ W(x,0) = \phi(x) - xh(0) \quad \text{(initial condition at } t = 0). \end{cases} \tag{46}$$

This new problem (46) has a homogeneous boundary condition to which we can apply the method of reflection or even extensions.

Diffusion with a source on a half-line (Cont'd)

For the inhomogeneous Neumann problem on the half line

$$\begin{aligned} &w_t - kw_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &w_x(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &w(x,0) = \phi(x) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{45}$$

- Let $W(x,t) = w(x,t) - xh(t)$, we can check that $W(x,t)$ satisfies

$$\begin{aligned} &W_t - kW_{xx} = f(x,t) - xh'(t), \quad 0 < x < \infty, \quad t > 0, \\ &W_x(0,t) = 0 \quad \text{(boundary condition at } x = 0), \\ &W(x,0) = \phi(x) - xh(0) \quad \text{(initial condition at } t = 0). \end{aligned} \tag{46}$$

  This new problem (46) has a homogeneous boundary condition to which we can apply the method of reflection or even extensions.
- $w(x,t)$ can be recovered by $w(x,t) = W(x,t) + xh(t)$ once $W(x,t)$ is found.

§3.4 Waves with a source

Waves with a source on the whole line

Consider the inhomogeneous wave equation

$$\begin{aligned} &u_{tt} - c^2 u_{xx} = f(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \end{aligned} \tag{47}$$

where $f(x)$, $\phi(x)$, and $\psi(x)$ are arbitrary functions.

§3.4 Waves with a source

Waves with a source on the whole line

Consider the inhomogeneous wave equation

$$\begin{aligned} &u_{tt} - c^2 u_{xx} = f(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \end{aligned} \tag{47}$$

where $f(x)$, $\phi(x)$, and $\psi(x)$ are arbitrary functions.

- Similar to the inhomogeneous heat equation, the right hand side of the equation, $f(x,t)$, is called the source term.

§3.4 Waves with a source

Waves with a source on the whole line

Consider the inhomogeneous wave equation

$$\begin{aligned} &u_{tt} - c^2 u_{xx} = f(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \end{aligned} \tag{47}$$

where $f(x)$, $\phi(x)$, and $\psi(x)$ are arbitrary functions.

- Similar to the inhomogeneous heat equation, the right hand side of the equation, $f(x,t)$, is called the source term.
- In the case of the string vibrations this source term measures the external force (per unit mass) applied on the string, and the equation again arises from Newton's second law, in which one now has a nonzero external force.

§3.4 Waves with a source

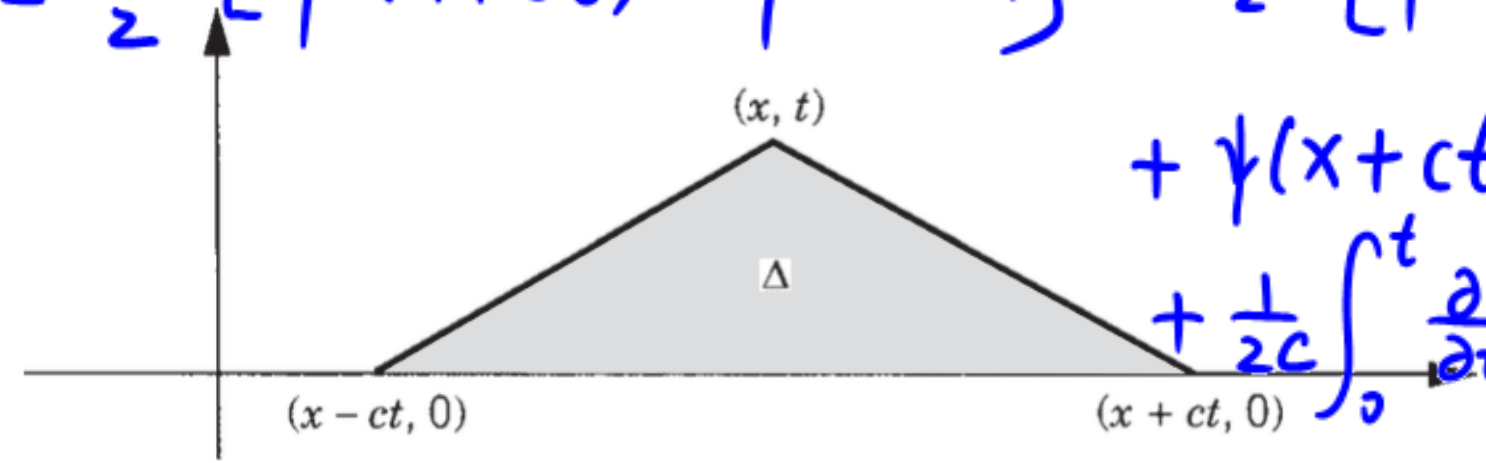
$u(x,0) = \frac{1}{2}[\phi(x) + \phi(x)] = \phi(x)$

Theorem 1. The unique solution of (47) is,

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds. \tag{48}$$

The iterated integral in (48) is over the characteristic triangle (see below).

$u_t = \frac{c}{2}[\phi'(x+ct) - \phi'(x-ct)] + \frac{1}{2}[\psi(x+ct) + \psi(x+ct)] + \frac{1}{2c}\int_0^t \frac{\partial}{\partial t}\int_{x-c(t-s)}^{x+c(t-s)}$



Proof of Theorem 1. We integrate f over the past history triangle (characteristic triangle) Δ. Thus

$$\iint_{\Delta} f dxdt = \iint_{\Delta} (u_{tt} - c^2 u_{xx}) dxdt. \tag{49}$$

Proof of Theorem 1. We integrate f over the past history triangle (characteristic triangle) Δ. Thus

$$\iint_{\Delta} f dxdt = \iint_{\Delta} (u_{tt} - c^2 u_{xx}) dxdt. \tag{49}$$

But Green's theorem says that

$$\iint_{\Delta} (P_x - Q_t) dxdt = \int_{\text{boundary}} P dt + Q dx \tag{50}$$

for any functions P and Q, where the line integral on the boundary is taken counterclockwise.

Proof of Theorem 1. We integrate f over the past history triangle (characteristic triangle) Δ. Thus

$$\iint_{\Delta} f dxdt = \iint_{\Delta} (u_{tt} - c^2 u_{xx}) dxdt. \tag{49}$$

But Green's theorem says that

$$\iint_{\Delta} (P_x - Q_t) dxdt = \int_{\text{boundary}} P dt + Q dx \tag{50}$$

for any functions P and Q, where the line integral on the boundary is taken counterclockwise.

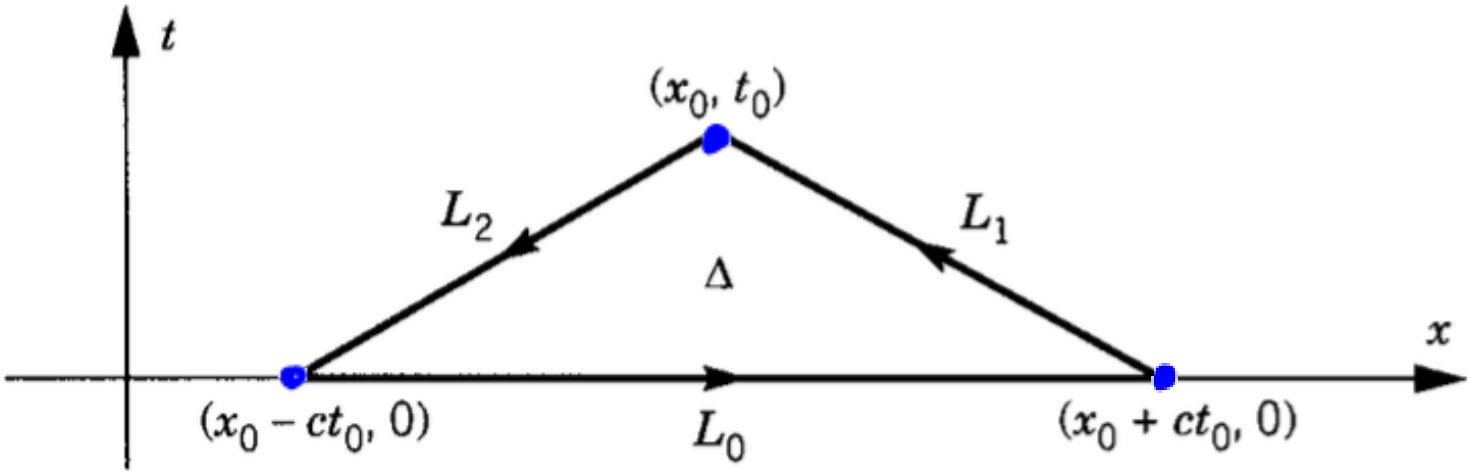
We assume that $P = -c^2 u_x$ and $Q = -u_t$, then

$$P_x - Q_t = -c^2 u_{xx} + u_{tt}. \tag{51}$$

Proof of Theorem 1 (Cont'd). Thus (49) can be rewritten as

$$\iint_{\Delta} f dx dt = \int_{L_0+L_1+L_2} (-c^2 u_x dt - u_t dx). \tag{52}$$

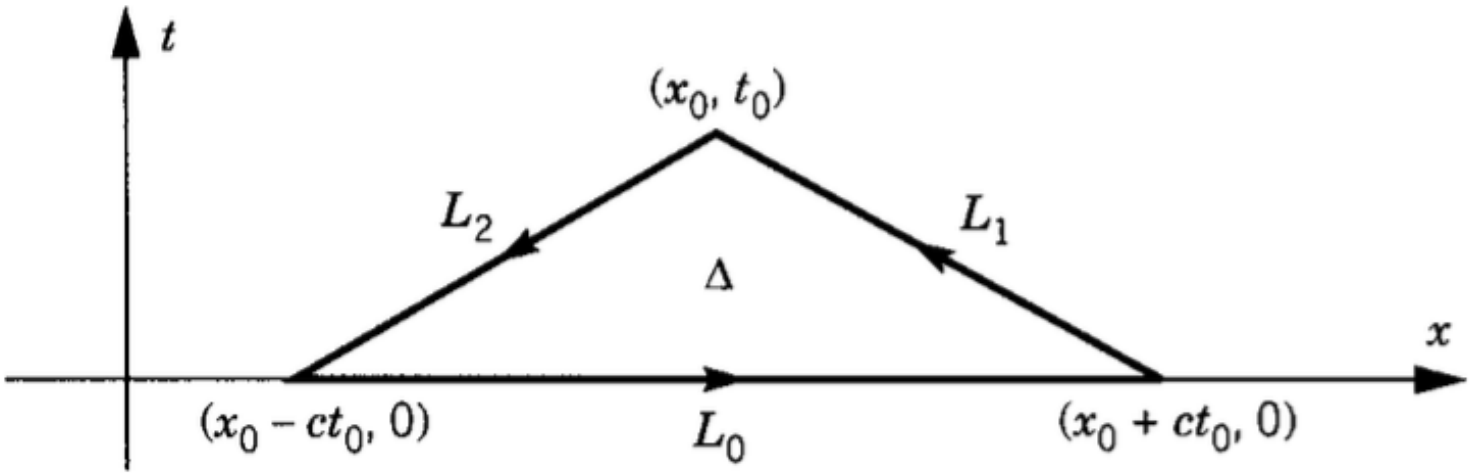
This is the sum of three line integrals over straight line segments (see Figure below). We evaluate each piece separately.



Proof of Theorem 1 (Cont'd). On L_0,

$t = 0$, $dt = 0$, and $u_t(x,t) = u_t(x,0) = \psi(x)$, so that

$$\int_{L_0} (-c^2 u_x dt - u_t dx) = \int_{L_0} (-u_t dx) = -\int_{x_0 - ct_0}^{x_0 + ct_0} \psi(x) dx. \tag{53}$$



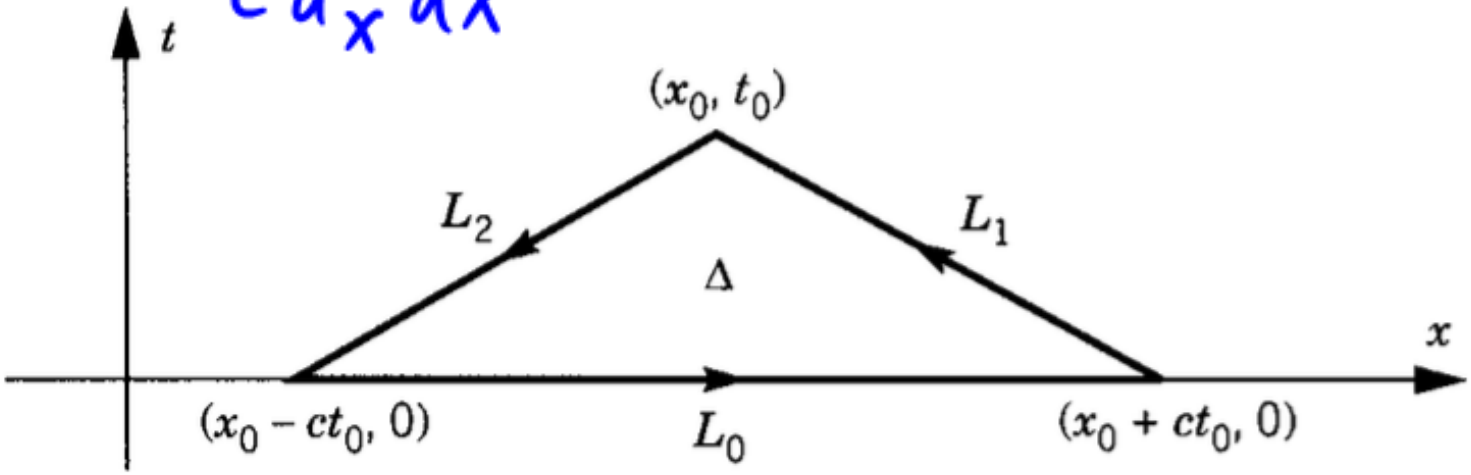
Proof of Theorem 1 (Cont'd). On L_1,

$x + ct = x_0 + ct_0 \longrightarrow dx + cdt = 0$ or $dt = -dx/c$ or $dx = -cdt$. Thus

$$\begin{aligned}\int_{L_1} (-c^2 u_x dt - u_t dx) &= \int_{L_1} (cu_x dx + cu_t dt) = c \int_{L_1} du \\ &= cu(x_0, t_0) - cu(x_0 + ct_0, 0) \\ &= cu(x_0, t_0) - c\phi(x_0 + ct_0).\end{aligned} \tag{54}$$

$dx = -c\,dt$

$cu_x\,dx$

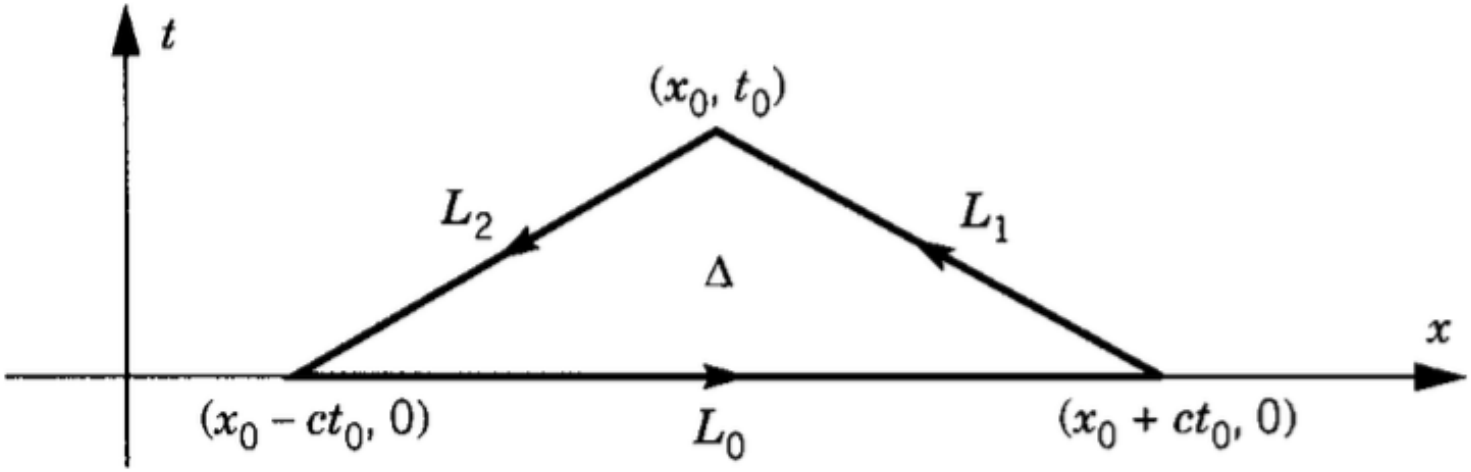


Proof of Theorem 1 (Cont'd). On L_2,

$x - ct = x_0 - ct_0 \longrightarrow dx - cdt = 0$ or $dt = dx/c$ or $dx = cdt$. Thus

$$\begin{aligned}\int_{L_2}(-c^2 u_x dt - u_t dx) &= \int_{L_2}(-cu_x dx - cu_t dt) = -c\int_{L_2} du \\ &= -cu(x_0 - ct_0, 0) + cu(x_0, t_0) \\ &= cu(x_0, t_0) - c\phi(x_0 - ct_0).\end{aligned} \tag{55}$$

$dx = c\,dt$



Proof of Theorem 1 (Cont'd).

(52) can be calculated by adding up (53), (54), and (55) as

$$\begin{aligned}\iint_{\Delta} f dxdt = &-\int_{x_0-ct_0}^{x_0+ct_0} \psi(x)dx + cu(x_0,t_0) - c\phi(x_0+ct_0) \\ &+ cu(x_0,t_0) - c\phi(x_0-ct_0).\end{aligned} \tag{56}$$

Proof of Theorem 1 (Cont'd).

(52) can be calculated by adding up (53), (54), and (55) as

$$\begin{aligned}\iint_{\Delta} f dxdt = &-\int_{x_0-ct_0}^{x_0+ct_0} \psi(x)dx + cu(x_0,t_0) - c\phi(x_0+ct_0) \\ &+ cu(x_0,t_0) - c\phi(x_0-ct_0).\end{aligned} \tag{56}$$

Thus

$$\begin{aligned}u(x_0,t_0) = &\frac{1}{2}\left[\phi(x_0-ct_0)+\phi(x_0+ct_0)\right] + \frac{1}{2c}\int_{x_0-ct_0}^{x_0+ct_0}\psi(x)dx \\ &+ \frac{1}{2c}\iint_{\Delta} f dxdt.\end{aligned} \tag{57}$$

Proof of Theorem 1 (Cont'd).

(52) can be calculated by adding up (53), (54), and (55) as

$$\begin{aligned}\iint_{\Delta} f dxdt = &-\int_{x_0-ct_0}^{x_0+ct_0} \psi(x)dx + cu(x_0,t_0) - c\phi(x_0+ct_0) \\ &+ cu(x_0,t_0) - c\phi(x_0-ct_0).\end{aligned} \tag{56}$$

Thus

$$\begin{aligned}u(x_0,t_0) = &\frac{1}{2}\left[\phi(x_0-ct_0)+\phi(x_0+ct_0)\right] + \frac{1}{2c}\int_{x_0-ct_0}^{x_0+ct_0}\psi(x)dx \\ &+ \frac{1}{2c}\iint_{\Delta} f dxdt.\end{aligned} \tag{57}$$

(x_0, t_0) can be any point in the half plane $t \geq 0$. (57) completes the proof.

§3.4 Waves with a source

Example B (not in the textbook)

Solve the inhomogeneous wave initial value problem,

$$\begin{aligned} &u_{tt} - c^2 u_{xx} = \delta(t-2)\cos x, \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = u_t(x,0) = 0, \qquad t > 0. \end{aligned} \tag{58}$$

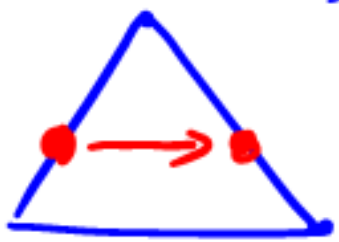
Example B (not in the textbook)

Solve the inhomogeneous wave initial value problem,

$$\begin{aligned} &u_{tt} - c^2 u_{xx} = \delta(t-2)\cos x, \quad -\infty < x < \infty, \quad t > 0, \\ &u(x,0) = u_t(x,0) = 0, \qquad t > 0. \end{aligned} \tag{58}$$

Solution. Using formula (48), and substituting the expression for $f(x,t) = \delta(t-2)\cos x$, $\phi(x) = 0$, and $\psi(x) = 0$, we get

(x,t)



$$\begin{aligned} u(x,t) &= \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \delta(s-2)\cos y dy ds \\ &= \frac{1}{2c}\int_0^t [\sin(x+c(t-s)) - \sin(x-c(t-s))]\delta(s-2) ds \\ &= \frac{1}{2c}\int_0^t 2\cos x \sin[c(t-s)]\delta(s-2) ds. \end{aligned}$$

Solution (Cont'd). Notice that if $t < 2$, then $\delta(s-2) = 0$ for all $s \in [0, t]$; and if $t > 2$, then the delta function will act on $2\cos x \sin[c(t-s)]$ by assigning its value at $s = 2$.

Solution (Cont'd). Notice that if $t < 2$, then $\delta(s-2) = 0$ for all $s \in [0, t]$; and if $t > 2$, then the delta function will act on $2\cos x \sin[c(t-s)]$ by assigning its value at $s = 2$.

Hence,

$$u(x,t) = \begin{cases} 0, & \text{for } \ 0 < t < 2, \\ \frac{1}{c}\cos x \sin[c(t-2)], & \text{for } \ t > 2. \end{cases}$$

Waves with a source on a half-line

Consider the Dirichlet problem of a boundary source $h(t)$ on the half-line; that is,

$$\begin{cases} v_{tt} - c^2 v_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ v(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ v(x,0) = \phi(x), \quad v_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0). \end{cases} \tag{59}$$

IBVP

§3.4 Waves with a source

Waves with a source on a half-line

Consider the Dirichlet problem of a boundary source $h(t)$ on the half-line; that is,

$$\begin{aligned} &v_{tt} - c^2 v_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &v(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &v(x,0) = \phi(x), \quad v_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0). \end{aligned} \tag{59}$$

- One can employ the subtraction method that we used for the heat equation to reduce the problem to one with zero Dirichlet data, and then use the reflection method to derive a solution formula for the reduced problem.

§3.4 Waves with a source

Waves with a source on a half-line

Consider the Dirichlet problem of a boundary source $h(t)$ on the half-line; that is,

$$\begin{aligned} &v_{tt} - c^2 v_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &v(0,t) = h(t) \quad \text{(boundary condition at } x = 0), \\ &v(x,0) = \phi(x), \quad v_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0). \end{aligned} \tag{59}$$

- One can employ the subtraction method that we used for the heat equation to reduce the problem to one with zero Dirichlet data, and then use the reflection method to derive a solution formula for the reduced problem.
- Let

$$V(x,t) = v(x,t) - h(t). \tag{60}$$

Waves with a source on a half-line (Cont'd)

- We can check that $V(x,t) = v(x,t) - h(t)$ satisfies

IBVP

$$\begin{cases} V_{tt} - c^2 V_{xx} = f(x,t) - h''(t), \quad 0 < x < \infty, \quad t > 0, \\ V(0,t) = 0, \\ V(x,0) = \phi(x) - h(0), \quad V_t(x,0) = \psi(x) - h'(0). \end{cases} \tag{61}$$

This new problem (61) has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions.

Waves with a source on a half-line (Cont'd)

- We can check that $V(x,t) = v(x,t) - h(t)$ satisfies

$$\begin{aligned} &V_{tt} - c^2 V_{xx} = f(x,t) - h''(t), \quad 0 < x < \infty, \quad t > 0, \\ &V(0,t) = 0, \\ &V(x,0) = \phi(x) - h(0), \quad V_t(x,0) = \psi(x) - h'(0). \end{aligned} \tag{61}$$

  This new problem (61) has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions.

- For simplicity, let $F(x,t) = f(x,t) - h''(t)$, $\Phi(x) = \phi(x) - h(0)$, and $\Psi(x) = \psi(x) - h'(0)$.

Waves with a source on a half-line (Cont'd)

- We can check that $V(x,t) = v(x,t) - h(t)$ satisfies

$$\begin{aligned} &V_{tt} - c^2 V_{xx} = f(x,t) - h''(t), \quad 0 < x < \infty, \quad t > 0, \\ &V(0,t) = 0, \\ &V(x,0) = \phi(x) - h(0), \quad V_t(x,0) = \psi(x) - h'(0). \end{aligned} \tag{61}$$

  This new problem (61) has a homogeneous boundary condition to which we can apply the method of reflection or odd extensions.

- For simplicity, let $F(x,t) = f(x,t) - h''(t)$, $\Phi(x) = \phi(x) - h(0)$, and $\Psi(x) = \psi(x) - h'(0)$.
- Then (61) becomes

$$\begin{aligned} &V_{tt} - c^2 V_{xx} = F(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &V(0,t) = 0, \\ &V(x,0) = \Phi(x), \quad V_t(x,0) = \Psi(x). \end{aligned} \tag{62}$$

Waves with a source on a half-line (Cont'd)

- Take the odd extensions:

$$\tilde{F}(x,t) = \begin{cases} F(x,t), & x > 0, \\ -F(-x,t), & x < 0, \\ 0, & x = 0, \end{cases} \quad \tilde{\Phi}(x) = \begin{cases} \Phi(x), & x > 0, \\ -\Phi(-x), & x < 0, \\ 0, & x = 0, \end{cases}$$
$$\tilde{\Psi}(x) = \begin{cases} \Psi(x), & x > 0, \\ -\Psi(-x), & x < 0, \\ 0, & x = 0. \end{cases} \tag{63}$$

Waves with a source on a half-line (Cont'd)

- Take the odd extensions:

$$\tilde{F}(x,t) = \begin{cases} F(x,t), & x > 0, \\ -F(-x,t), & x < 0, \\ 0, & x = 0, \end{cases} \quad \tilde{\Phi}(x) = \begin{cases} \Phi(x), & x > 0, \\ -\Phi(-x), & x < 0, \\ 0, & x = 0, \end{cases}$$
$$\tilde{\Psi}(x) = \begin{cases} \Psi(x), & x > 0, \\ -\Psi(-x), & x < 0, \\ 0, & x = 0. \end{cases} \tag{63}$$

- Then we solve the extended wave equation on the whole real axis:

$$\begin{aligned} &U_{tt} - c^2 U_{xx} = \tilde{F}(x,t), \quad -\infty < x < \infty, \quad t > 0, \\ &U(x,0) = \tilde{\Phi}(x), \quad U_t(x,0) = \tilde{\Psi}(x), \quad -\infty < x < \infty, \end{aligned} \tag{64}$$

Waves with a source on a half-line (Cont'd)

Using the solution formula (48), we have the solution to the extended wave equation (64) on the whole real axis; that is

$$U(x,t) = \frac{1}{2}[\tilde{\Phi}(x+ct) + \tilde{\Phi}(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\Psi}(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \tilde{F}(y,s)dyds. \tag{65}$$

Waves with a source on a half-line (Cont'd)

Using the solution formula (48), we have the solution to the extended wave equation (64) on the whole real axis; that is

$$U(x,t) = \frac{1}{2}[\tilde{\Phi}(x+ct) + \tilde{\Phi}(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\Psi}(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \tilde{F}(y,s)dyds. \tag{65}$$

We only consider the solution $U(x,t) = V(x,t) = v(x,t) - h(t)$ on the positive axis $x > 0$. Two different cases should be discussed.

Waves with a source on a half-line (Cont'd)

Using the solution formula (48), we have the solution to the extended wave equation (64) on the whole real axis; that is

$$U(x,t) = \frac{1}{2}[\tilde{\Phi}(x+ct) + \tilde{\Phi}(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \tilde{\Psi}(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} \tilde{F}(y,s)dyds. \tag{65}$$

We only consider the solution $U(x,t) = V(x,t) = v(x,t) - h(t)$ on the positive axis $x > 0$. Two different cases should be discussed.

(I) If $x > ct$ $(x > 0)$, then

$$V(x,t) = \frac{1}{2}[\Phi(x+ct) + \Phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \Psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} F(y,s)dyds. \tag{66}$$

Waves with a source on a half-line (Cont'd)

Since $V(x,t) = v(x,t) - h(t)$, (66) can be rewritten as

$$
\begin{aligned}
v(x,t) - h(t) &= \frac{1}{2}[\phi(x+ct) - h(0) + \phi(x-ct) - h(0)] \\
&\quad + \frac{1}{2c}\int_{x-ct}^{x+ct} [\psi(y) - h'(0)]\, dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} [f(y,s) - h''(s)]\, dyds \\
&= \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds \\
&\quad - h(0) - h'(0)t - \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} h''(s)dyds \\
&= \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds \\
&\quad - h(t).
\end{aligned}
\tag{67}
$$

Waves with a source on a half-line (Cont'd)

(II) If $x < ct$ $(x > 0)$, then

$$
\begin{aligned}
V(x,t) &= \frac{1}{2}[\Phi(x+ct) + \tilde{\Phi}(x-ct)] + \frac{1}{2c}\int_0^{x+ct} \Psi(y)dy + \frac{1}{2c}\int_{x-ct}^0 \tilde{\Psi}(y)dy \\
&\quad + \frac{1}{2c}\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} F(y,s)dyds + \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} F(y,s)dyds \\
&\quad + \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_{x-c(t-s)}^{0} \tilde{F}(y,s)dyds \\
&= \frac{1}{2}[\Phi(x+ct) - \Phi(ct-x)] + \frac{1}{2c}\int_0^{x+ct} \Psi(y)dy - \frac{1}{2c}\int_{x-ct}^0 \Psi(-y)dy \\
&\quad + \frac{1}{2c}\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} F(y,s)dyds + \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)} F(y,s)dyds \\
&\quad - \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_{x-c(t-s)}^{0} F(-y,s)dyds.
\end{aligned}
$$

Waves with a source on a half-line (Cont'd)

Make the change of variables in the blue integrals as y to $-y$, we get

$$
\begin{aligned}
V(x,t) &= \frac{1}{2}[\Phi(x+ct)-\Phi(ct-x)] + \frac{1}{2c}\int_0^{x+ct}\Psi(y)dy + \frac{1}{2c}\int_{-x+ct}^{0}\Psi(y)dy \\
&\quad + \frac{1}{2c}\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)}F(y,s)dyds + \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_0^{x+c(t-s)}F(y,s)dyds \\
&\quad + \frac{1}{2c}\int_0^{t-\frac{x}{c}}\int_{-x+c(t-s)}^{0}F(y,s)dyds \\
&= \frac{1}{2}[\Phi(x+ct)-\Phi(ct-x)] + \frac{1}{2c}\int_{ct-x}^{x+ct}\Psi(y)dy \\
&\quad + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\}F(y,s)dyds
\end{aligned}
\tag{69}
$$

Waves with a source on a half-line (Cont'd)

Substituting $F(x,t) = f(x,t) - h''(t)$, $\Phi(x) = \phi(x) - h(0)$, and $\Psi(x) = \psi(x) - h'(0)$ into (69) gives

$$\begin{aligned}
V(x,t) &= \frac{1}{2}[\phi(x+ct) - h(0) - \phi(ct-x) + h(0)] + \frac{1}{2c}\int_{ct-x}^{x+ct} [\psi(y) - h'(0)]\, dy \\
&\quad + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_{0}^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\}[f(y,s) - h''(s)]\, dyds \\
&= \frac{1}{2}[\phi(x+ct) - \phi(ct-x)] + \frac{1}{2c}\int_{ct-x}^{x+ct}\psi(y)dy - \frac{1}{2c}\int_{ct-x}^{x+ct} h'(0)dy \\
&\quad + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_{0}^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)dyds \\
&\quad - \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_{0}^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} h''(s)dyds
\end{aligned} \tag{70}$$

Waves with a source on a half-line (Cont'd)

(70) can be further simplified as

$$V(x,t) = \frac{1}{2}[\phi(x+ct) - \phi(ct-x)] + \frac{1}{2c}\int_{ct-x}^{x+ct} \psi(y)dy$$
$$+ \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_{0}^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)dyds \tag{71}$$
$$-h(t) + h(t-\frac{x}{c}).$$

Waves with a source on a half-line (Cont'd)

(70) can be further simplified as

$$V(x,t) = \frac{1}{2}[\phi(x+ct) - \phi(ct-x)] + \frac{1}{2c}\int_{ct-x}^{x+ct} \psi(y)dy$$
$$+ \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_{0}^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)dyds \tag{71}$$
$$-h(t) + h(t-\frac{x}{c}).$$

Since $V(x,t) = v(x,t) - h(t)$, when $x < ct$ $(x > 0)$ we have

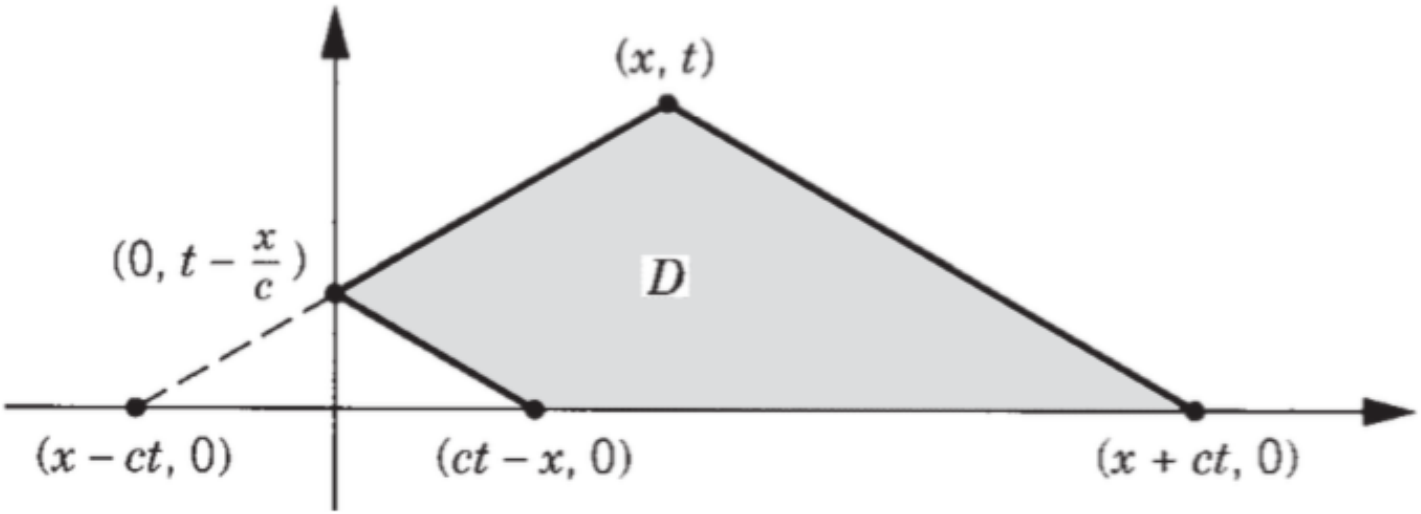
$$v(x,t) = h(t-\frac{x}{c}) + \frac{1}{2}[\phi(x+ct) - \phi(ct-x)] + \frac{1}{2c}\int_{ct-x}^{x+ct} \psi(y)dy$$
$$+ \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_{0}^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)dyds. \tag{72}$$

Waves with a source on a half-line (Cont'd)

The sum of the two double integrals

$$\frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_{0}^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)dyds. \tag{73}$$

is over the dependence domain of (x,t).



Waves with a source on a half-line (Cont'd)

Therefore, the solution to the Dirichlet problem (59) of a boundary source $h(t)$ on the half-line ($x > 0$) is

when $x > ct$

$$v(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)dy + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dyds. \tag{74}$$

when $x < ct$

$$v(x,t) = h(t - \frac{x}{c}) + \frac{1}{2}[\phi(x+ct) - \phi(ct-x)] + \frac{1}{2c}\int_{ct-x}^{x+ct} \psi(y)dy + \frac{1}{2c}\left\{\int_{t-\frac{x}{c}}^{t}\int_{x-c(t-s)}^{x+c(t-s)} + \int_0^{t-\frac{x}{c}}\int_{c(t-s)-x}^{x+c(t-s)}\right\} f(y,s)dyds. \tag{75}$$

Waves with a source on a half-line

Consider the Neumann problem of a boundary source $g(t)$ on the half-line; that is,

$$\begin{aligned} &w_{tt} - c^2 w_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &w_x(0,t) = g(t) \quad \text{(boundary condition at } x = 0), \\ &w(x,0) = \phi(x), \quad w_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0). \end{aligned} \tag{76}$$

§3.4 Waves with a source

Waves with a source on a half-line

Consider the Neumann problem of a boundary source $g(t)$ on the half-line; that is,

$$\begin{aligned} &w_{tt} - c^2 w_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &w_x(0,t) = g(t) \quad \text{(boundary condition at } x = 0), \\ &w(x,0) = \phi(x), \quad w_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0). \end{aligned} \tag{76}$$

- We can also use the subtraction method to reduce the problem to one with zero Neumann data.

§3.4 Waves with a source

Waves with a source on a half-line

Consider the Neumann problem of a boundary source $g(t)$ on the half-line; that is,

$$\begin{aligned} &w_{tt} - c^2 w_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &w_x(0,t) = g(t) \quad \text{(boundary condition at } x = 0), \\ &w(x,0) = \phi(x), \quad w_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0). \end{aligned} \tag{76}$$

- We can also use the subtraction method to reduce the problem to one with zero Neumann data.
- Let

$$W(x,t) = w(x,t) - xg(t). \tag{77}$$

§3.4 Waves with a source

Waves with a source on a half-line

Consider the Neumann problem of a boundary source $g(t)$ on the half-line; that is,

$$\begin{aligned} &w_{tt} - c^2 w_{xx} = f(x,t), \quad 0 < x < \infty, \quad t > 0, \\ &w_x(0,t) = g(t) \quad \text{(boundary condition at } x = 0), \\ &w(x,0) = \phi(x), \quad w_t(x,0) = \psi(x), \quad \text{(initial condition at } t = 0). \end{aligned} \tag{76}$$

- We can also use the subtraction method to reduce the problem to one with zero Neumann data.
- Let
$$W(x,t) = w(x,t) - xg(t). \tag{77}$$
- Then use the reflection method to derive a solution formula for the reduced problem.

MH4110 PDE

Tutorial 06

Question 1

Consider the heat equation

$$\frac{\partial u}{\partial t} - 4\frac{\partial^2 u}{\partial x^2} = 0$$

(a) Show that it is linear.

(b) Show that it is parabolic.

(c) If $x^2 + at + bx$ is a solution, find the values of a and b.

(d) If $u(x, 0) = x^2 + 2x$, find $u(x, t)$.

(a) To check the linearity of an operator $\mathscr{L}$, we only need check if the following two equations are satisfied or not

$$\mathscr{L}(u + v) = \mathscr{L}u + \mathscr{L}v \qquad \mathscr{L}(ku) = k\mathscr{L}u.$$

Here u and v are the solutions to the discussing PDE.

We have

$$\mathscr{L}(u + v) = (u + v)_t - 4(u + v)_{xx} = (u_t - 4u_{xx}) + (v_t - 4v_{xx}) = \mathscr{L}u + \mathscr{L}v,$$

and

$$\mathscr{L}(ku) = (ku)_t - 4(ku)_{xx} = k\,(u_t - 4u_{xx}) = k\mathscr{L}u.$$

So $\mathscr{L}$ here is a linear operator.

Question 1

Consider the heat equation

$$\frac{\partial u}{\partial t} - 4\frac{\partial^2 u}{\partial x^2} = 0$$

(a) Show that it is linear.

(b) Show that it is parabolic.

(c) If $x^2 + at + bx$ is a solution, find the values of a and b.

(d) If $u(x, 0) = x^2 + 2x$, find $u(x, t)$.

(b) The coefficient matrix is

$$A = \begin{bmatrix} 0 & 0 \\ 0 & -4 \end{bmatrix}.$$

Its determinant is $\det(A) = 0$. The equation is of parabolic type.

(c) Give that $u(x, t) = x^2 + at + bx$, we have $u_t = a$ and $u_{xx} = 2$. Since $0 = u_t - 4u_{xx}$, $a - 8 = 0$ or $a = 8$. b can be any arbitrary real number.

(d) From the solution formula we know that

$$u(x, t) = \int_{-\infty}^{\infty} S(x - y, t)\phi(y)dy = \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-y)^2}{4kt}} (y^2 + 2y)dy.$$

Exercise

Evaluate

$$u(x,t) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-y)^2}{4kt}} (y^2 + 2y) dy.$$

Question 2

Consider the PDE

$$u_x + 2u_y = 2x$$

(a) Find the general solution to this PDE,

(b) Find the solution to the PDE that additionally satisfies $u(x, 0) = 5x^2$.

(a) We know that the general solution of the inhomogeneous problem is the sum of the general solution to the homogeneous problem $u_x + 2u_y = 0$ and a particular solution to the inhomogeneous problem. It is not difficult for us to find that $u(x, y) = x^2$ is a particular solution because of the specific and simple form of the inhomogeneous term $2x$ on the right hand side. Using the method of characteristics, we know that the general solution to the homogeneous problem $u_x + 2u_y = 0$ is $u(x, y) = f(2x - y)$ with f as an arbitrary function. So the general solution to the original problem is

$$u(x, y) = f(2x - y) + x^2.$$

(b) Given that $u(x, 0) = 5x^2$, we have $5x^2 = u(x, 0) = f(2x) + x^2$, indicating that $f(x) = x^2$. So the final solution is

$$u(x, y) = 5x^2 - 4xy + y^2.$$

Question 2

Consider the PDE

$$u_x + 2u_y = 2x$$

(a) Find the general solution to this PDE,

(b) Find the solution to the PDE that additionally satisfies $u(x, 0) = 5x^2$.

(a) We alternatively make the change of variables to find the solution. Change variables to

$$x' = x + 2y \qquad y' = 2x - y$$

Replace all x and y derivatives by x' and y' derivatives. By the chain rule,

$$u_x = \frac{\partial u}{\partial x} = \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial x} = u_{x'} + 2u_{y'}$$

and

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial y} = 2u_{x'} - u_{y'}.$$

Hence $\frac{2}{5}(x' + 2y') = 2x = u_x + 2u_y = (u_{x'} + 2u_{y'}) + 2(2u_{x'} - u_{y'}) = 5u_{x'}$. The equation takes the form $u_{x'} = \frac{2}{25}(x' + 2y')$ in the new (primed) variables. Thus the solution is $u = \frac{x'^2}{25} + \frac{4x'y'}{25} + g(y')$, with g an arbitrary function. The general solution in the original coordinates is

$$u(x, y) = \frac{(x + 2y)^2}{25} + \frac{4(x + 2y)(2x - y)}{25} + g(2x - y) = \frac{9x^2 + 16xy - 4y^2}{25} + g(2x - y).$$

Question 3

Find the general solution to this PDE

$$u_x + xu_y = 0.$$

We look for the characteristic curve $y = y(x)$ satisfying

$$\frac{dy}{dx} = x \quad \Rightarrow \quad y - \frac{1}{2}x^2 = C.$$

Hence, the general solution of the problem is

$$u(x, y) = f(y - \frac{1}{2}x^2).$$

with f as an arbitrary function.

Question 4

Consider the PDE

$$u_x + 2xu_y = xy^3.$$

(a) Find the general solution to this PDE,

(b) Find the solution to the PDE that additionally satisfies $u(x, 0) = x^4$.

(a) The general solution of the inhomogeneous problem is the sum of the general solution to the homogeneous problem and a particular solution to the inhomogeneous problem. For the homogeneous problem, we look for the characteristic curve $y = y(x)$ satisfying

$$\frac{dy}{dx} = 2x \quad \Rightarrow \quad y - x^2 = C.$$

Hence, the general solution of the homogeneous problem is $u(x, y) = f(y - x^2)$ with f as an arbitrary function.
We assume that the particular solution is constant in y. That means $u(x, y) = g(x)$ and $g'(x) = xy^3$. But we cannot find such a function of g. We alternatively assume that the particular solution is constant in x. That means $u(x, y) = g(y)$ and $2xg'(y) = xy^3$. Luckily, we have many options of g. For simplicity, g can be chosen as $g(y) = y^4/8$. So, $u(x, y) = y^4/8$ is a particular solution. Thus, the solution to the inhomogeneous problem is $u(x, y) = f(y - x^2) + \frac{y^4}{8}$.

(b) Since $u(x, 0) = x^4$, we have $u(x, 0) = f(-x^2) = x^4$, indicating that $f(x) = x^2$. So, the explicit form of the solution satisfying the additional condition $u(x, 0) = x^4$ is $u(x, y) = (x^2 - y)^2 + \frac{y^4}{8}$.

Question 5

Let $u(t,x)$ be the solution of the following IVP for the heat equation

$$\begin{cases} u_t - ku_{xx} = 0, & -\infty < x < \infty, \quad t \in R \\ u(x,0) = \phi(x), & -\infty < x < \infty. \end{cases}$$

Show that if ϕ is an odd function of x, then $u(-x,t) = -u(x,t)$ for all x, $t \in R$.

From the solution formula we know that

$$u(x,t) = \int_{-\infty}^{\infty} S(x-y,t)\phi(y)dy = \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-y)^2}{4kt}} \phi(y)dy.$$

Since $\phi(x)$ is an odd function, $\phi(-x) = -\phi(x)$, the solution can be rewritten as

$$u(x,t) = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-y)^2}{4kt}} \phi(-y)dy.$$

Let $p = -y$, we have

$$u(x,t) = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x+p)^2}{4kt}} \phi(p)dp.$$

Using $-x$ to replace x in the above integral gives that

$$u(-x,t) = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(-x+p)^2}{4kt}} \phi(p)dp = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-p)^2}{4kt}} \phi(p)dp = -u(x,t).$$

Therefore, we get that $u(-x,t) = -u(x,t)$, meaning that $u(x,t)$ is an odd function.

Question 6

Find the solution $u(t,x)$ to the following IVP for the heat equation

$$\begin{cases} u_t - ku_{xx} = 0, & -\infty < x < \infty, \quad t > 0 \\ u(x,0) = \phi(x), & -\infty < x < \infty. \end{cases}$$

where $\phi(x) = 3$ for $0 < x < 2$ and $\phi(x) = 0$ otherwise. Determine the maximum and minimum values of $u(t,x)$ on the domain $R = \{(t,x) \in R^2 : -\infty < x < \infty, 0 \le t < \infty\}$.

We have the solution formula

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} \exp\Big(-\frac{(x-y)^2}{4kt}\Big)\phi(y)dy.$$

Substituting the explicit form of $\phi(x)$ gives

$$u(x,t) = \frac{3}{\sqrt{4\pi kt}} \int_{0}^{2} \exp\Big(-\frac{(x-y)^2}{4kt}\Big)dy.$$

Based on the maximum principle, the maximum/minimum value can be only achieved at the bottom boundary (t=0) or the two lateral sides. Since $-\infty < x < \infty$, no lateral boundaries exist. The maximum/minimum values can be only assumed at $t = 0$. When $t = 0$, $u(x,t) = u(x,0) = \phi(x)$. Therefore, the maximum value is 3, and the minimum value is 0.

Question 7

Find the solution $u(t,x)$ to the following IBVP for the heat equation

$$\begin{cases} u_t - ku_{xx} = 0, & 0 < x < \infty, \quad t > 0 \\ u(x,0) = \phi(x), & 0 < x < \infty, \\ u(0,t) = 0, & t \in R. \end{cases}$$

where $\phi(x) = 3$ for $0 < x < 2$ and $\phi(x) = 0$ otherwise. Determine the maximum and minimum values of $u(t,x)$ on the domain $R = \{(t,x) \in R^2 : 0 \le x < \infty, 0 \le t < \infty\}$.

For $0 < x < \infty$ and $0 < t < \infty$, we have the solution formula for the Dirichlet IBVP as follows

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) - \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]\phi(y)dy.$$

Substituting the explicit form of $\phi(x)$ into the above expression gives

$$u(x,t) = \frac{3}{\sqrt{4\pi kt}} \int_0^2 \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) - \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]dy.$$

Based on the maximum principle, the maximum/minimum value can be only achieved at the bottom boundary ($t = 0$) or the two lateral sides. Since $0 \le x < \infty$, only the left lateral boundary exists, and hence the maximum/minimum values can be assumed at $t = 0$ or $x = 0$. When $x = 0$, $u(x,t) = u(0,t) = 0$; when $t = 0$, $u(x,t) = u(x,0) = \phi(x)$. Therefore, the maximum value is 3 (when $t = 0$), and the minimum value is 0 (when $t = 0$ or $x = 0$).

Question 8

Find the solution $u(t,x)$ to the following IBVP for the heat equation

$$\begin{cases} u_t - ku_{xx} = 0, & 0 < x < \infty, \quad t > 0 \\ u(x,0) = \phi(x), & 0 < x < \infty; \qquad u_x(0,t) = 0, \quad t \in R. \end{cases}$$

where $\phi(x) = 3$ for $0 < x < 2$ and $\phi(x) = 0$ otherwise. Determine the maximum and minimum values of $u(t,x)$ on the domain $R = \{(t,x) \in R^2 : 0 \le x < \infty, 0 \le t < \infty\}$.

For $0 < x < \infty$ and $0 < t < \infty$, we have the solution formula for the Neumann IBVP as follows

$$u(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_0^\infty \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) + \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]\phi(y)dy.$$

Substituting the explicit form of $\phi(x)$ into the above expression gives

$$u(x,t) = \frac{3}{\sqrt{4\pi kt}} \int_0^2 \Big[\exp\Big(-\frac{(x-y)^2}{4kt}\Big) + \exp\Big(-\frac{(x+y)^2}{4kt}\Big)\Big]dy.$$

Based on the maximum principle, the maximum/minimum value can be only achieved at the bottom boundary ($t = 0$) or the two lateral sides. When $x = 0$,

$$u(0,t) = \frac{6}{\sqrt{4\pi kt}} \int_0^2 \exp\Big(-\frac{y^2}{4kt}\Big)dy < \frac{6}{\sqrt{4\pi kt}} \int_0^\infty \exp\Big(-\frac{y^2}{4kt}\Big)dy = 6\int_0^\infty S(y,t)dy = 3$$

It is obvious that $u(0,t) > 0$. When $t = 0$, $u(x,t) = u(x,0) = \phi(x)$. Therefore, the maximum value is 3 (when $t = 0$), and the minimum value is 0 (when $t = 0$).

Question 9

Find the general solution $u(t,x)$ to the PDE

$$u_{xx} - 4u_{yy} = x^2, \quad -\infty < x < \infty, \quad -\infty < y < \infty.$$

The characteristic lines of the PDE are $y \pm 2x = C$ (C is an arbitrary constant). We can make the change of variables to find the solutions. Change variables to

$$x' = 2x + y \qquad y' = -2x + y$$

Replace all x and y derivatives by x' and y' derivatives. By the chain rule,

$$u_x = \frac{\partial u}{\partial x} = \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial x} = 2u_{x'} - 2u_{y'}$$

and

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial u}{\partial y'}\frac{\partial y'}{\partial y} = u_{x'} + u_{y'}.$$

Hence $\frac{x'^2 - 2x'y' + y'^2}{16} = x^2 = u_{xx} - 4u_{yy} = -16u_{x'y'}$. The equation takes the form $u_{x'y'} = \frac{-x'^2 + 2x'y' - y'^2}{256}$ in the new (primed) variables.

Question 9

The equation takes the form

$$u_{x'y'} = \frac{-x'^2 + 2x'y' - y'^2}{256}$$

in the new (primed) variables.

First integrating with respect to y' gives

$$u_{x'} = \frac{-x'^2y' + x'y'^2 - \frac{1}{3}y'^3}{256} + f(x').$$

f is an arbitrary function.

Then integrating $u_{x'}$ with respect to x', we have

$$u = \frac{-\frac{1}{3}x'^3y' + \frac{1}{2}x'^2y'^2 - \frac{1}{3}x'y'^3}{256} + F(x') + G(y').$$

Again, F and G are arbitrary functions. Changing back to the original coordinates, the solution can be written as

$$u = \frac{112x^4 - 24x^2y^2 - y^4}{1536} + F(2x + y) + G(-2x + y).$$

Question 10

Find the solution $u(t,x)$ to the following IVP for the wave equation

$$\begin{cases} u_{tt} - 4u_{xx} = 0, & -\infty < x < \infty, \quad t > 0 \\ u(x,0) = x^2, \quad u_t(x,0) = 1, & -\infty < x < \infty. \end{cases}$$

The speed of the waves is $c = 2$. Using the d'Alembert's formula for the homogeneous initial value problem, we have

$$u(x,t) = \frac{1}{2}\left[(x+2t)^2 + (x-2t)^2\right] + \frac{1}{4}\int_{x-2t}^{x+2t} ds.$$

Therefore, the solution is

$$u(x,t) = x^2 + 4t^2 + t.$$

Question 11

Let $u(t,x)$ be the solution of the following initial value problem for the wave equation

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0, & -\infty < x < \infty, \quad t > t_0 \\ u(x,t_0) = \phi(x), \quad u_t(x,t_0) = \psi(x), & -\infty < x < \infty. \end{cases}$$

If $\phi(x)$ and $\psi(x)$ are even functions of x, show that the solution $u(x,t)$ of the wave equation is also even in x.

Let $\tau = t - t_0$, then

$$\begin{cases} u_{\tau\tau} - c^2 u_{xx} = 0, & -\infty < x < \infty, \quad \tau \geq 0 \\ u(x,\tau)|_{\tau=0} = \phi(x), \quad u_\tau(x,\tau)|_{\tau=0} = \psi(x), & -\infty < x < \infty. \end{cases}$$

By the d'Alembert formula,

$$u(x,\tau) = \frac{\phi(x-c\tau) + \phi(x+c\tau)}{2} + \frac{1}{2c}\int_{x-c\tau}^{x+c\tau} \psi(s)ds.$$

or

$$u(x,t) = \frac{\phi(x-c(t-t_0)) + \phi(x+c(t-t_0))}{2} + \frac{1}{2c}\int_{x-c(t-t_0)}^{x+c(t-t_0)} \psi(s)ds.$$

Question 11

By the d'Alembert formula,

$$u(x,\tau) = \frac{\phi(x-c\tau)+\phi(x+c\tau)}{2} + \frac{1}{2c}\int_{x-c\tau}^{x+c\tau} \psi(s)ds.$$

or

$$u(x,t) = \frac{\phi(x-c(t-t_0))+\phi(x+c(t-t_0))}{2} + \frac{1}{2c}\int_{x-c(t-t_0)}^{x+c(t-t_0)} \psi(s)ds.$$

Since $\phi(-s)=\phi(s)$ and $\psi(-s)=\psi(s)$, we have

$$u(-x,t) = \frac{\phi(-x-c(t-t_0))+\phi(-x+c(t-t_0))}{2} + \frac{1}{2c}\int_{-x-c(t-t_0)}^{-x+c(t-t_0)} \psi(s)ds.$$

Make a change of variables as $s=-y$ and note that $ds=-dy$, then

$$\begin{aligned} u(-x,t) &= \frac{\phi(x+c(t-t_0))+\phi(x-c(t-t_0))}{2} - \frac{1}{2c}\int_{x+c(t-t_0)}^{x-c(t-t_0)} \psi(-y)dy. \\ &= \frac{\phi(x+c(t-t_0))+\phi(x-c(t-t_0))}{2} + \frac{1}{2c}\int_{x-c(t-t_0)}^{x+c(t-t_0)} \psi(-y)dy \\ &= u(x,t). \end{aligned}$$

This means $u(x,t)$ is even in x for all t.

Question 12

Find the solution $u(t,x)$ to the following IBVP for the wave equation

$$\begin{cases} u_{tt} - 4u_{xx} = 0, & 0 < x < \infty, \quad t > 0 \\ u(x,0) = x^2, \quad u_t(x,0) = 1, & 0 < x < \infty \\ u(0,t) = 0, & t > 0. \end{cases}$$

In the lecture we have derived the solution formula for the Dirichlet IBVP

$$u(x,t) = \begin{cases} \frac{1}{2}\left[\phi(x+ct) + \phi(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds, & x > ct, \\ \\ \frac{1}{2}\left[\phi(x+ct) - \phi(ct-x)\right] + \frac{1}{2c}\int_{ct-x}^{ct+x} \psi(s)ds, & 0 < x < ct. \end{cases}$$

For this IBVP, we have $c = 2$, $\phi(x) = x^2$, and $\psi(x) = 1$. The solution is therefore

$$u(x,t) = \begin{cases} \frac{1}{2}\left[(x+2t)^2 + (x-2t)^2\right] + \frac{1}{4}\int_{x-2t}^{x+2t} ds, & x > 2t, \\ \\ \frac{1}{2}\left[(x+2t)^2 - (2t-x)^2\right] + \frac{1}{4}\int_{2t-x}^{2t+x} ds, & 0 < x < 2t. \end{cases}$$

After simplification, we have

$$u(x,t) = \begin{cases} x^2 + 4t^2 + t, & x > 2t, \\ \\ 4tx + \frac{1}{2}x, & 0 < x < 2t. \end{cases}$$

Question 13

Find the solution $u(t,x)$ to the following IBVP for the wave equation

$$\begin{cases} u_{tt} - 4u_{xx} = 0, & 0 < x < \infty, \quad t > 0 \\ u(x,0) = x^2, \quad u_t(x,0) = 1, & 0 < x < \infty \\ u_x(0,t) = 0, & t > 0. \end{cases}$$

We have the solution formula for the Neumann problem on the half-line (**??**):

$$u(x,t) = \begin{cases} \dfrac{\phi(x+ct) + \phi(x-ct)}{2} + \dfrac{1}{2c}\displaystyle\int_{x-ct}^{x+ct} \psi(s)ds, & x > ct, \\ \\ \dfrac{\phi(x+ct) + \phi(ct-x)}{2} + \dfrac{1}{2c}\left(\displaystyle\int_0^{ct-x} + \int_0^{x+ct}\right)\psi(s)ds, & 0 < x < ct. \end{cases}$$

For this IBVP, we have $c = 2$, $\phi(x) = x^2$, and $\psi(x) = 1$. The solution is therefore

$$u(x,t) = \begin{cases} \dfrac{(x+2t)^2 + (x-2t)^2}{2} + \dfrac{1}{4}\displaystyle\int_{x-2t}^{x+2t} 1ds, & x > 2t, \\ \\ \dfrac{(x+2t)^2 + (2t-x)^2}{2} + \dfrac{1}{4}\left(\displaystyle\int_0^{2t-x} + \int_0^{x+2t}\right)1ds, & 0 < x < 2t. \end{cases}$$

Question 13

For this IBVP, we have $c = 2$, $\phi(x) = x^2$, and $\psi(x) = 1$. The solution is therefore

$$u(x,t) = \begin{cases} \dfrac{(x+2t)^2 + (x-2t)^2}{2} + \dfrac{1}{4}\displaystyle\int_{x-2t}^{x+2t} 1ds, & x > 2t, \\ \\ \dfrac{(x+2t)^2 + (2t-x)^2}{2} + \dfrac{1}{4}\left(\displaystyle\int_0^{2t-x} + \int_0^{x+2t}\right) 1ds, & 0 < x < 2t. \end{cases}$$

After simplification, we have

$$u(x,t) = \begin{cases} x^2 + 4t^2 + t, & x > 2t, \\ \\ x^2 + 4t^2 + t, & 0 < x < 2t. \end{cases}$$

So,

$$u(x,t) = x^2 + 4t^2 + t, \quad x > 0.$$

Question 14

Use the method of factoring the operator to find the general solution $u(t,x)$ to the following PDE $u_{xx} + 6u_{xy} - 16u_{yy} = 16x + 2y$.

The general solution of the inhomogeneous problem is the sum of the general solution to the homogeneous problem $u_{xx} + 6u_{xy} - 16u_{yy} = 0$ and a particular solution to the inhomogeneous problem. Let's first find a particular solution to the inhomogeneous problem. Considering that the inhomogeneous term on the right hand side is a polynomial of degree 1, the differential operator on the left hand side is second order, we can guess a particular solution can be a polynomial of degree 3. Thus, we assume that a particular solution has the expression as

$$u(x,y) = ax^3 + by^3,$$

with undetermined coefficients a and b. A direct substitution into the inhomogeneous PDE gives that

$$6ax - 96by = 16x + 2y,$$

indicating that $a = 8/3$ and $b = -1/48$. So, we have a particular solution

$$u(x,y) = \frac{8}{3}x^3 - \frac{1}{48}y^3.$$

We now use the method of factoring the operator to solve the homogeneous problem. Observe that

$$u_{xx} + 6u_{xy} - 16u_{yy} = \Big(\frac{\partial}{\partial x} - 2\frac{\partial}{\partial y}\Big)\Big(\frac{\partial}{\partial x} + 8\frac{\partial}{\partial x}\Big)u. \tag{1}$$

Define

$$v = \Big(\frac{\partial}{\partial x} - 2\frac{\partial}{\partial y}\Big)u = u_x - 2u_y. \tag{2}$$

Then (1) can be rewritten as

$$v_x + 8v_y = 0. \tag{3}$$

Therefore, we can solve (3) for finding v, and then find u by solving (2).
Applying the method of characteristics to (3) leads to

$$v(x,t) = h(8x - y), \tag{4}$$

where h is an arbitrary function.

Plugging (4) into (2) gives

$$u_x - 2u_y = h(8x - y). \tag{5}$$

It is a linear equation, whose general solution is a linear combination of a particular solution and the general solution to the homogeneous equation: $u_x - 2u_y = 0$ as in ODE. We know the general solution of $u_x - 2u_y = 0$ is $g(2x + y)$, where g is an arbitrary function. In order to find a particular solution, we try this solution:

$$u = f(8x - y) \quad \Rightarrow \quad u_x - 2u_y = 8f'(8x - y) + 2f'(8x - y) = h(8x - y),$$

where f' is the ordinary derivative of a function of one variable. We see that if we take $f'(s) = h(s)/10$, then $u = f(8x - y)$ is a particular solution of (5). Finally, we obtain the solution of $u_{xx} + 2u_{xy} - 16u_{yy} = 0$ as $u(x,y) = f(8x - y) + g(2x + y)$, where f, g are arbitrary functions. Therefore, the general solution to the original problem is

$$u(x,y) = f(8x - y) + g(2x + y) + \frac{8}{3}x^3 - \frac{1}{48}y^3.$$

Question 15

Classify each of the following PDE as hyperbolic, elliptic, or parabolic

(a) $u_{xx} + 2u_{xy} + u_{yy} + u_x + u_y = 0$,

(b) $u_{xx} + 2u_{xy} + 2u_{yy} + u_x + u_y = \sin(xy)$,

(c) $2u_{xx} - 4u_{xy} - 6u_{yy} + u_x = 0$.

(a) We see that $a_{11} = 1, a_{12} = 1, a_{22} = 1$, so we have $a_{11}a_{22} - a_{12}^2 = 0$. The equation is of parabolic type.

(b) The coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its determinant is $\det(A) = 1 \times 2 - 1 \times 1 > 0$. The equation is of elliptic type.

(c) The coefficient matrix is

$$A = \begin{bmatrix} 2 & -2 \\ -2 & -6 \end{bmatrix}.$$

Its determinant is $\det(A) = 2 \times (-6) - (-2) \times (-2) = -16 < 0$. The equation is of hyperbolic type.